Predicting Growth and Remodeling of Cardiovascular Tissue

change in mass

change in microstructure

Dr.-Ing. Martin R. Pfaller

Assistant Professor

Department of Biomedical Engineering, Yale University

biomechanics.yale.edu

With much inspiration from Drs. Amadeus Gebauer, Jonas Eichinger, and Jay Humphrey

Student Version

Yale

The Origin of Mechanobiology

Davis' law, **Henry Gassett Davis** (1807–1896, USA), 1867:

"Ligaments, or any soft tissue, when put under even a moderate degree of tension, if that tension is unremitting, will elongate by the addition of new material."

Wolff's law, **Julius Wolff** (1836–1902, Germany), 1892 (*Das Gesetz der Transformation der Knochen*, the law of bone transformation):

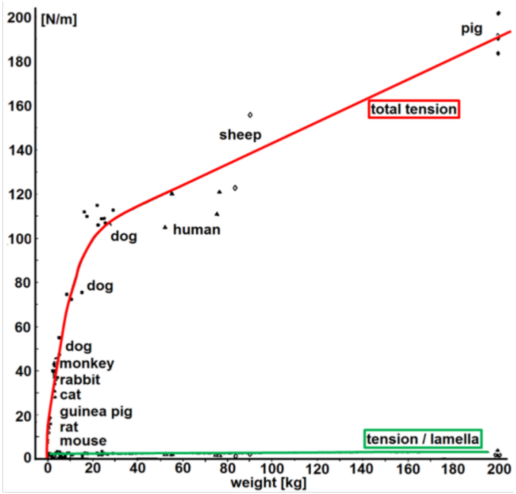
"im Gefolge primärer Abänderungen der Form und Inanspruchnahme oder auch bloß der Inanspruchnahme der Knochen [sich] bestimmte, nach mathematischen Regeln eintretende Umwandlungen der inneren Architektur und ebenso bestimmte, denselben mathematischen Regeln folgende sekundäre Umwandlungen der äußeren Form der betreffenden Knochen vollziehen."

- bone mass proportional to load (higher / lower load $\Rightarrow$ growth / atrophy)
- first *mathematical* relation between load and growth & remodeling in biological tissue

Davis (1867); Wolff, Das Gesetz der Transformation der Knochen (1892)

A Brief Introduction to Mechanobiology of Blood Vessels

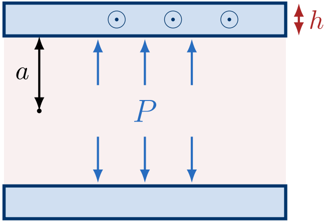
Vascular tissue has a preferred mechanical state



Tension T varies by a **factor of 26**

$$T = P\,a$$

T and σ_θ

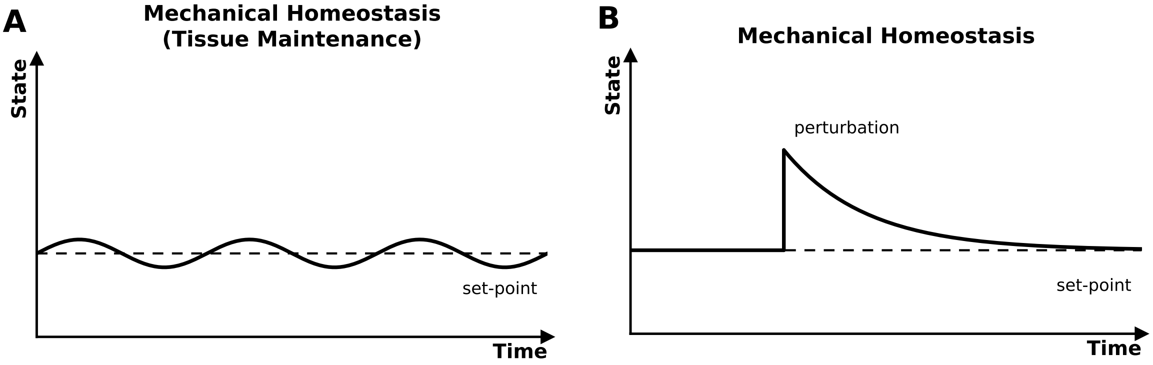


Tension / lamella varies by a **factor of 2.8**

$$\frac{T}{h} = \frac{P\,a}{h} = \sigma_\theta$$

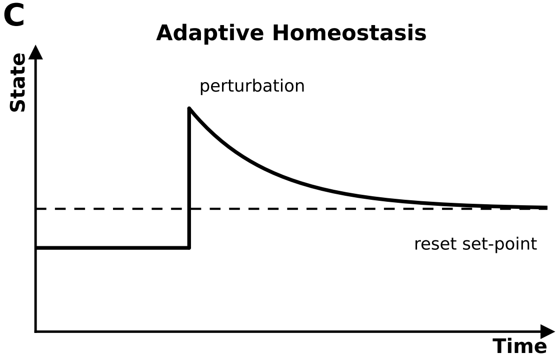
Wolinsky & Glagov, Circ Res (1967); Fig. 8 adapted by J. Eichinger

Mechanical Homeostasis



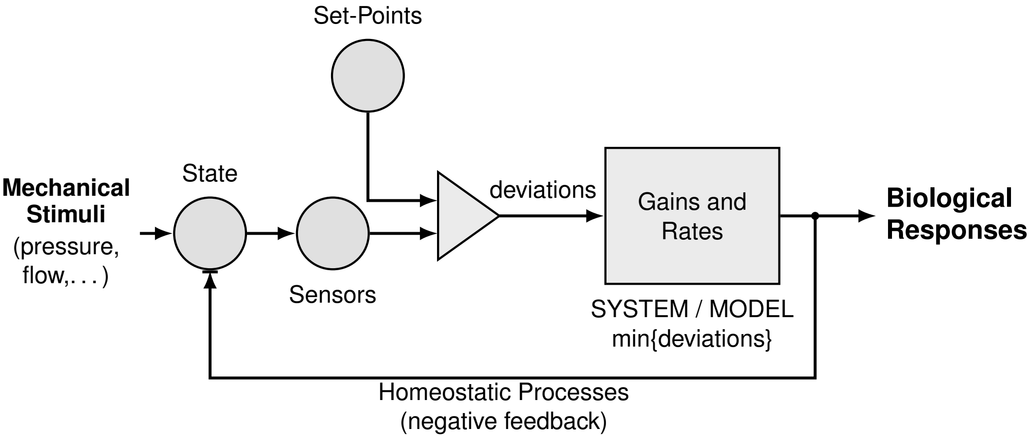
Humphrey & Schwartz, Annu Rev Biomed Eng (2021)

(Mal)adaptation



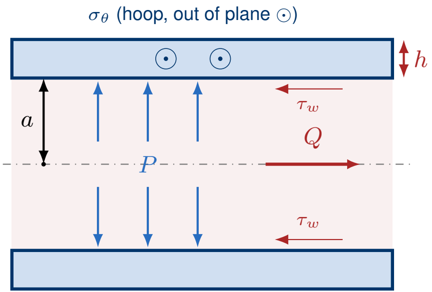
Humphrey & Schwartz, Annu Rev Biomed Eng (2021)

Mechanobiological Regulation



An Idealized Blood Vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure** P pushes it outward, and **blood flow** Q drags its inner surface. Cells sense both.



a inner radius, h wall thickness
P lumen pressure $\to$ **circumferential stress** $\sigma_\theta \sim 100\,$kPa
Q volumetric flow $\to$ **wall shear stress** $\tau_w \sim 1\,$Pa

Humphrey, Cardiovascular Solid Mechanics (2002)

Two Loads, Two Formulae, Two Adaptations

Pressure P sets the **hoop stress**; flow Q sets the **wall shear stress**. Each has a set-point; rearranging shows the geometry that restores it.

Pressure (Laplace)

$$\sigma_\theta = \frac{Pa}{h} \tag{1}$$

Flow (Poiseuille)

$$\tau_w = \frac{4\mu Q}{\pi\, a^3} \tag{2}$$

Your turn: Pressure adaptation

Question. Pressure **doubles**. How does the wall thickness change? $\sigma_\theta = \frac{Pa}{h}$

Your turn: Pressure adaptation

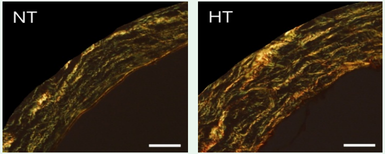
Question. Pressure **doubles**. How does the wall thickness change? $\sigma_\theta = \dfrac{Pa}{h}$

Answer: Pressure adaptation

$P'/h' = P/h$ with $P' = 2P$ gives

$$h' = \left(\frac{P'}{P}\right) h = 2\,h. \tag{3}$$

Wall thickens *in proportion* to pressure.



$P \uparrow$

Your turn: Flow adaptation

Question. Flow rises **8-fold**. How does the radius change? $\tau_w = \dfrac{4\mu Q}{\pi\, a^3}$

Your turn: Flow adaptation

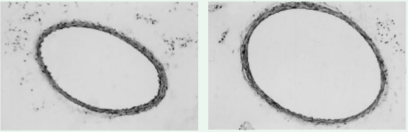
Question. Flow rises **8-fold**. How does the radius change? $\tau_w = \dfrac{4\mu Q}{\pi\, a^3}$

Answer: Flow adaptation

$(a'/a)^3 = Q'/Q = 8$, so

$$a' = \left(\frac{Q'}{Q}\right)^{1/3} a = 2\,a. \qquad (4)$$

Reading. Radius grows as the *cube root* of flow.



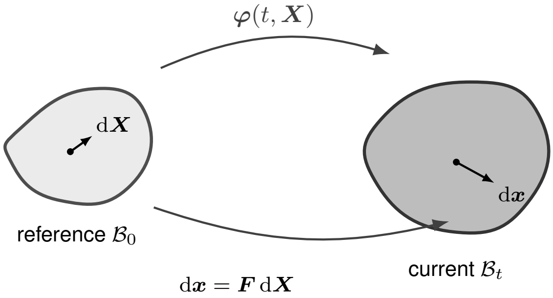
$Q\uparrow$

Finite-Strain Fundamentals

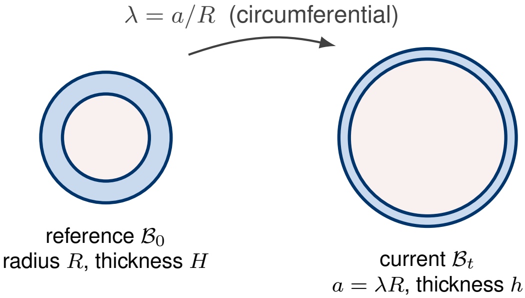
Reference and Current Configuration

A body is described by a **reference** $\mathcal{B}_0$ (points $\boldsymbol{X}$) and a **current** $\mathcal{B}_t$ (points $\boldsymbol{x}$) configuration, related by the motion $\boldsymbol{x} = \boldsymbol{\varphi}(t, \boldsymbol{X})$.

3D 3D geometric model



0D 0D reduced model



The Deformation Gradient

The **deformation gradient** maps reference line elements to current ones.

3D a 3×3 tensor

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}, \qquad \boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}}, \qquad J = \det \boldsymbol{F} \tag{5}$$

J is the volume ratio, **incompressibility** means $J = 1$.

0D a scalar

$$\boldsymbol{F} \;\longrightarrow\; \lambda = \frac{\text{current length}}{\text{reference length}}, \qquad \lambda = 1 \text{ undeformed.}$$

Strain: A Rotation-Free Measure of Stretch

Removing rigid rotation gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**.

3D symmetric 3×3 tensors

$$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = (\boldsymbol{R}\boldsymbol{U})^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{U} = \boldsymbol{U}^2, \qquad \boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{6}$$

0D scalars

$$C = \lambda^2, \qquad E = \tfrac{1}{2}(\lambda^2 - 1).$$

Stress: "Force per Area"

Cauchy stress: *current* force per unit *current* area.

3D spatial symmetric 3×3 tensor (from 2nd Piola–Kirchhoff stress $\boldsymbol{S}$)

$$\boldsymbol{\sigma} = \tfrac{1}{J}\,\boldsymbol{F}\,\boldsymbol{S}\,\boldsymbol{F}^{\mathrm{T}}, \qquad \boldsymbol{S} = 2\,\frac{\partial W}{\partial \boldsymbol{C}}, \quad W \text{ strain energy density function.}$$

0D scalar reduction ($C = \lambda^2$, $J = 1$), the σ_θ in Laplace (1)

$$\sigma = \tfrac{1}{J}\,\lambda\,S\,\lambda = \lambda^2 S = 2\lambda^2\,\frac{\partial W}{\partial C} = \lambda\,\frac{\mathrm{d}W}{\mathrm{d}\lambda}.$$

Hyperelasticity: Stress from a Stored-Energy Function

A **hyperelastic** material stores energy W (deformation is path-independent).

3D energy $W(\boldsymbol{C})$, stress $\boldsymbol{S} = 2\,\partial W/\partial \boldsymbol{C}$

$$W_{\text{neo-Hooke}} = \tfrac{c}{2}\,(I_1 - 3), \qquad W_{\text{Fung}} = \tfrac{c_1}{4c_2}\left[e^{\,c_2(I_4-1)^2} - 1\right]$$

Invariants of $\boldsymbol{C}$: $I_1 = \operatorname{tr}\boldsymbol{C}$ (isotropic, elastin); $I_4 = \boldsymbol{M}\cdot\boldsymbol{C}\,\boldsymbol{M} = \lambda_{\text{fib}}^2$, the squared stretch along the reference fiber direction $\boldsymbol{M}$ (anisotropy, collagen).

0D reduction used here, one stretch λ, scalar $\sigma = \lambda\,\mathrm{d}W/\mathrm{d}\lambda$

$$\sigma_{\text{neo-Hooke}} = c\,(\lambda^2 - \lambda^{-1}), \qquad \sigma_{\text{Fung}} = c_1\,\lambda^2(\lambda^2 - 1)\,e^{\,c_2(\lambda^2-1)^2} \tag{7}$$

Kinematic Growth

Multiplicative Decomposition: Growth vs. Elasticity

Split the deformation into a stress-free **growth** part $\boldsymbol{F}_g$ and a stress-carying **elastic** part $\boldsymbol{F}_e$.

3D a product of tensors

$$\boldsymbol{F} = \boldsymbol{F}_e\,\boldsymbol{F}_g, \qquad \text{only } \boldsymbol{F}_e \text{ stores energy.}$$

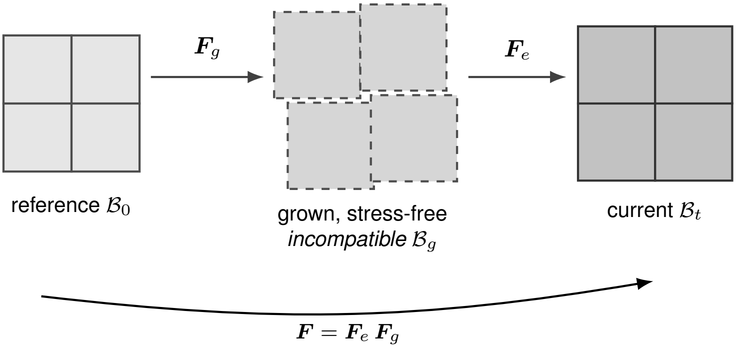
0D scalar stretches

$$\lambda = \lambda_e\,\lambda_g \tag{8}$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Multiplicative Decomposition: Growth vs. Elasticity

$\boldsymbol{F}_g$ yields an *incompatible* stress-free state, $\boldsymbol{F}_e$ restores compability via residual stress.



Kinematic Growth Tensor

$\boldsymbol{F}_g$ encodes *how* the tissue is assumed to add mass. A common choice grows the tissue an amount θ along a fixed direction $\boldsymbol{f}$:

3D prescribed growth tensor

$$\boldsymbol{F}_g = \boldsymbol{I} + (\theta - 1)\,\boldsymbol{f} \otimes \boldsymbol{f} \;=\; \mathrm{diag}(1,\,1,\,\theta) \tag{9}$$

0D growth stretch

$$\lambda_g \quad \text{(scalar growth stretch)}, \qquad \theta = \det \boldsymbol{F}_g = M/M_0.$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Growth Law

The growth *pattern* $\boldsymbol{F}_g$ is typically fixed. Its rate $\dot{\boldsymbol{F}}_g = \dot{\theta}\, \boldsymbol{f} \otimes \boldsymbol{f}$ reduces to *one* scalar ODE:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = k_g\, \theta \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right) \tag{10}$$

Since $k_g\, \theta > 0$, the sign of the response follows the stress deviation:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} \begin{cases} > 0 & \bar{\sigma} > \bar{\sigma}_h \quad \text{(growth)} \\ = 0 & \bar{\sigma} = \bar{\sigma}_h \quad \text{(homeostasis)} \\ < 0 & \bar{\sigma} < \bar{\sigma}_h \quad \text{(atrophy)} \end{cases}$$

Rodriguez, Hoger, McCulloch, J Biomech (1994)

Getting Started with the Code

One runner drives *every* exercise: you never edit Python, just point it at a small input file. **First-time setup** (needs Python ≥ 3.10; dependencies: NumPy, SciPy, Matplotlib, PyYAML):

```
git clone https://github.com/YaleCBL-Teaching/Growth-Remodeling
cd Growth-Remodeling
```

uv (recommended):

```
uv sync
```

pip (your own Python):

```
python -m venv .venv
pip install -e .
```

What is where:

- `run.py`: the single runner for all exercises
- `configs/`: one YAML input file per exercise *(you edit these)*
- `src/gr/`: the models & maths (read, don't edit)
- `docs/`: documentation
- `slides/`: semianr notes

Setup: Example (hypertensive artery, `ex01`)

Example. A pressurised artery is made *hypertensive* by a sustained pressure step. Kinematic growth thickens the wall. Does the wall stress $\bar{\sigma}/\bar{\sigma}_h$ return to the *prescribed* set-point?

Running an Exercise

Point the runner at the input file (`uv run`, or plain `python` once the venv is activated):

```
uv run python run.py configs/ex01_kinematic_growth.yaml
```

You get: the key numbers ($\bar{\sigma}/\bar{\sigma}_h$, mass, `diverged`) print to the terminal, and the time history opens as a plot.

Change one value: edit the YAML, or override on the command line:

```
uv run python run.py configs/ex01_kinematic_growth.yaml -set insult.pressure_factor=2
```

Parametric study: one curve per value:

```
...  -sweep insult.pressure_factor=1.5,2,3
```

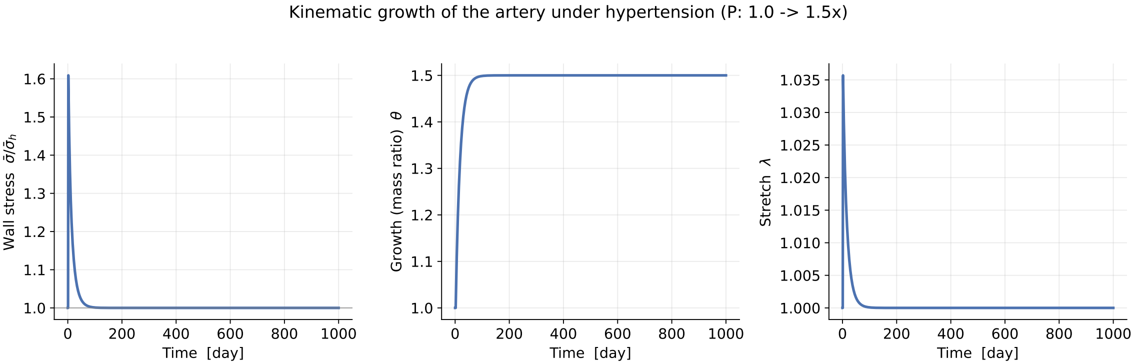
Your turn: Kinematic growth

Question. Does the wall stress return to the prescribed set-point? Try larger pressure steps. Does the outcome ever change?

Hints. $\bar{\sigma}/\bar{\sigma}_h$ settles at 1 while θ climbs to the pressure factor. Under pure hypertension it *always* adapts. Runaway needs the mixture / aneurysm story (later).

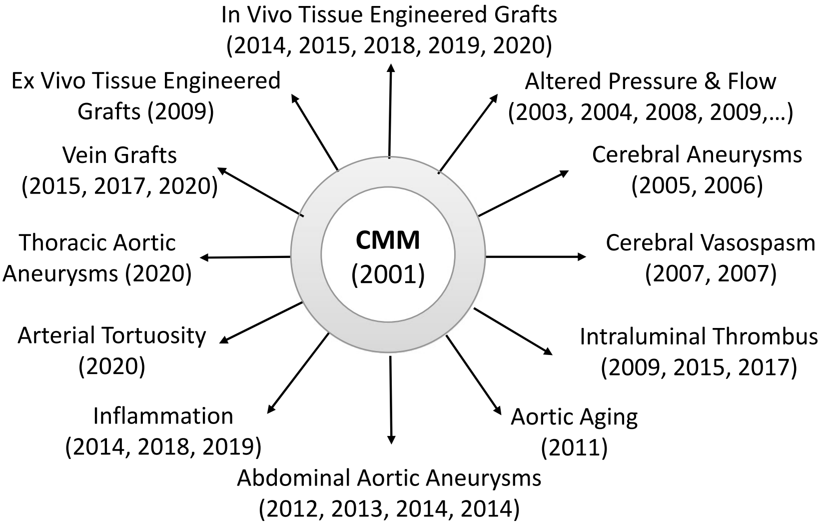
Answer: Kinematic growth

The stress plateau *always* lands on the prescribed set-point. The wall simply thickens more for a larger pressure step. Kinematic growth *imposes* stability; it cannot predict runaway.



The Constrained Mixture Model

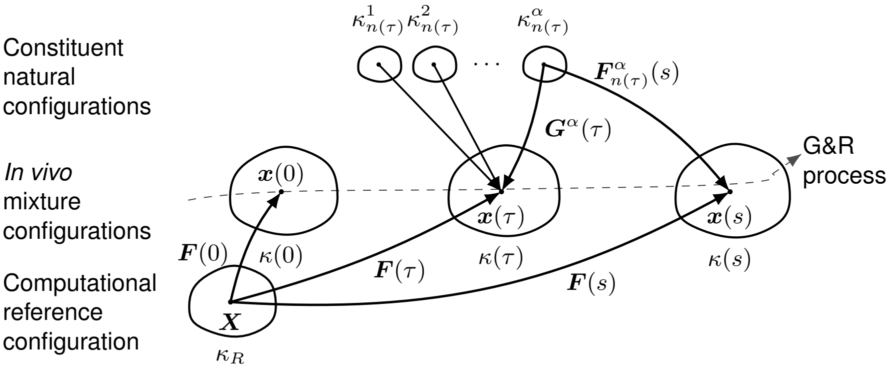
One Framework, Many Applications



Ubiquitous Vascular G&R

Humphrey, J Elasticity (2021)

The Idea: Track Constituents Over Time



κ: configuration • τ: deposition time • s: current time • α: constituent • $\boldsymbol{G}^\alpha$: deposition stretch • $\boldsymbol{F}^\alpha_{n(\tau)}$: constituent deformation

Humphrey, J Elasticity (2021)

Load-Bearing Vessel Constituents

The wall is a mixture of three constituents, each with a reference **mass fraction** ϕ_0^α ($\sum_\alpha \phi_0^\alpha = 1$) and a **deposition stretch** G^α, the prestretch cells lay it down at, which *defines* its homeostatic stress $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; the mixture set-point is $\bar{\sigma}_h = \sum_\alpha \phi_0^\alpha \sigma_h^\alpha$.

Constituent	Role	ϕ_0	G^α	law; turnover $1/k_d$
Elastin	bears load at low stretch	0.30	1.40	neo-Hooke; **none** (decades)
Collagen	stiff, crimped; anti-rupture	0.35	1.08	Fung; fast ($\sim$80 d)
Smooth muscle	passive + active; the sensor	0.35	1.20	Fung; fast ($\sim$80 d)

Elastin cannot be renewed ($k_d^e = 0$), its loss drives *aneurysm*; collagen & SMC **turn over**, so the wall can *adapt*.

Every Cohort Shares One Stretch

A cohort of α deposited at τ is laid down at G^{α} and then stretches *with the mixture*: every constituent and cohort is **constrained** to the same tissue stretch $\lambda(s)$.

3D constituent-specific deformation

$$\boldsymbol{F}^{\alpha}_{n(\tau)}(s) = \boldsymbol{F}(s)\,\boldsymbol{F}^{-1}(\tau)\,\boldsymbol{G}^{\alpha}(\tau).$$

0D scalar elastic stretch

$$\lambda^{\alpha}_{n(\tau)}(s,\tau) = \frac{\lambda(s)}{\lambda(\tau)}\,G^{\alpha} \tag{11}$$

Mass Balance

Track each constituent's **apparent mass density** ρ^α per current mixture volume, $\rho = \sum_\alpha \rho^\alpha$.

Spatial mass balance (one per constituent):

$$\frac{\partial \rho^\alpha}{\partial s} + \underbrace{\mathrm{div}(\rho^\alpha \boldsymbol{v}^\alpha)}_{\text{transport}} = \underbrace{\bar{m}^\alpha}_{\text{net mass exchange}}, \qquad \alpha = 1, \ldots, N.$$

The net rate is *signed*:

$$\bar{m}^\alpha = \underbrace{m^\alpha}_{\text{production}\geq 0} - \underbrace{r^\alpha}_{\text{removal}\geq 0} \begin{cases} < 0 & \text{atrophy} \\ = 0 & \text{maintenance} \\ > 0 & \text{growth} \end{cases}$$

From Balance to a Heredity Integral

Assumption 1 (constrained mixture): every constituent moves with the mixture, $\boldsymbol{v}^\alpha = \boldsymbol{v}$.

Assumption 2 (prescribed rates): a true production rate $m^\alpha(\tau) \geq 0$ and a survival fraction $q^\alpha(s, \tau) \in [0, 1]$. Integrating from the distant past to s:

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0)\, Q^\alpha(s)}_{\text{survivors of original material}} + \underbrace{\int_0^s m^\alpha(\tau)\, q^\alpha(s, \tau)\, \mathrm{d}\tau}_{\text{deposited at } \tau,\ \text{still present at } s} \tag{12}$$

with $Q^\alpha(0) = 1$. **Assumption 3:** bulk density is nearly constant, $\rho(s) = \rho(0)$, mass changes track volume changes.

Linear Momentum Balance

A rule-of-mixtures for the stresses (no body forces) satisfies the classical linear-momentum balance:

$$\operatorname{div} \boldsymbol{\sigma} = \rho\, \boldsymbol{a} \quad \xrightarrow{\text{quasi-static}} \quad \operatorname{div} \boldsymbol{\sigma} = \mathbf{0}.$$

Quasi-static assumption: G&R evolves slowly at every time s, so the inertia $\rho \boldsymbol{a}$ is negligible.

Constitutive Relations: Cauchy Stress

In finite elasticity the Cauchy stress follows from a stored-energy function W (per unit reference mixture volume):

$$\boldsymbol{\sigma}(s) = \frac{2}{J}\,\boldsymbol{F}(s)\,\frac{\partial W(s)}{\partial \boldsymbol{C}(s)}\,\boldsymbol{F}^{\mathrm{T}}(s), \qquad \boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}, \quad J = \det \boldsymbol{F}.$$

Rule of mixtures: $W = \sum_\alpha \hat{W}^\alpha$. Closing the model needs **three constituent-specific relations**:

- a *stored energy* W^α,
- a *production* rate m^α, and
- a *removal* (survival) q^α.

Three Constituent-Specific Constitutive Relations

1. Material: constituent stored energy W^α (e.g., neo-Hookean, Fung-type)
2. Rate of production: up with wall stress and inflammation, down with wall shear:

$$m^\alpha(\tau) = m_o^\alpha \big(1 + \underbrace{K_\sigma^\alpha \, \Delta\sigma}_{\text{circumferential stress}} - \underbrace{K_{\tau_w}^\alpha \, \Delta\tau_w}_{\text{wall shear stress}} + \underbrace{K_\varphi^\alpha \, \Delta\rho_\varphi}_{\text{inflammation}}\big) \geq 0 \tag{13}$$

3. Rate of removal: a survival fraction that decays faster under stress perturbation:

$$q^\alpha(s,\tau) = \exp\Big(-\int_\tau^s k_o^\alpha \big(1 + \omega \, \Delta\sigma^2(t) + \cdots\big) \, \mathrm{d}t\Big) \in [0,1] \tag{14}$$

Constituent Stored-Energy Functions

Each cohort carries its *own* stored energy W^α. Summing over cohorts with the *same* heredity structure gives the apparent (mixture-level) energy:

$$\rho\,\hat{W}^\alpha(s) = \underbrace{\rho^\alpha(0)\,Q^\alpha(s)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(s)\big)}_{\text{original material}} + \underbrace{\int_0^s m^\alpha(\tau)\,q^\alpha(s,\tau)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(\tau)}(s)\big)\,\mathrm{d}\tau}_{\text{later cohorts}} \tag{15}$$

, with elastic deformation

$$\boldsymbol{F}^\alpha_{n(\tau)}(s) = \boldsymbol{F}(s)\,\boldsymbol{F}^{-1}(\tau)\,\boldsymbol{G}^\alpha(\tau) \tag{16}$$

. The mixture energy $W = \sum_\alpha \hat{W}^\alpha$ closes the mixture stress.

Special Cases: Before Perturbation and Homeostasis

(I) At $s = 0$ (just before a perturbation) only original material is present, and $Q^\alpha(0) = 1$:

$$\rho\,\hat{W}^\alpha(0) = \rho^\alpha(0)\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(0)\big) \;\Rightarrow\; \hat{W}^\alpha(0) = \underbrace{\tfrac{\rho^\alpha(0)}{\rho(0)}}_{\phi^\alpha(0)}\,W^\alpha(0),$$

recovering the rule of mixtures $W(0) = \sum_\alpha \phi^\alpha(0)\,W^\alpha(0)$.

(II) Homeostasis: turnover in an *unchanging* configuration, so $\boldsymbol{F}^\alpha_{n(\tau)}(s) = \boldsymbol{F}^\alpha_{n(0)}(s)$ (independent of τ). Then W^α factors out and, using the mass balance,

$$\rho\,\hat{W}^\alpha(s) = \underbrace{\left(\rho^\alpha(0)Q^\alpha(s) + \int_0^s m^\alpha q^\alpha\,\mathrm{d}\tau\right)}_{=\,\rho^\alpha(s)}\,W^\alpha\big(\boldsymbol{F}^\alpha_{n(0)}(s)\big) \;\Rightarrow\; \hat{W}^\alpha(s) = \phi^\alpha(s)\,W^\alpha(s). \tag{17}$$

Perfect Maintenance

Take *constant* rates: $m^\alpha = m_0$, $Q^\alpha(s) = e^{-K^\alpha s}$, $q^\alpha(s,\tau) = e^{-K^\alpha (s-\tau)}$. The mass balance integrates to

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0)\, e^{-K^\alpha s}}_{\text{original decays}} + \underbrace{\frac{m_0}{K^\alpha}\left(1 - e^{-K^\alpha s}\right)}_{\text{production accumulates}} .$$

A time-*invariant* density ($\rho^\alpha(s) = \rho^\alpha(0)$ for all s) requires the two to balance exactly:

$$\text{perfect maintenance} \iff \frac{m_0}{K^\alpha} = \rho^\alpha(0). \tag{18}$$

Production m_0 then exactly offsets removal $K^\alpha \rho^\alpha$ (continuous turnover, unchanging tissue).

Your turn: Computational Cost?

Question. At each of N time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with N?

Hints. Step k re-sums $\sim k$ stored cohorts; add up $\sum_{k=1}^{N} k$.

Your turn: Computational Cost?

Question. At each of N time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with N?

Hints. Step k re-sums $\sim k$ stored cohorts; add up $\sum_{k=1}^{N} k$.

Answer: Computational Cost?

Step k re-sums the $\sim k$ cohorts stored so far, so the total work is

$$\sum_{k=1}^{N} k = \frac{N(N+1)}{2} = \mathcal{O}(N^2)$$

quadratic in the number of time steps. The *homogenized* model (next) removes this history sum, cutting the cost to $\mathcal{O}(N)$.

Setup: Example 2: Turnover under Hypertension (ex02)

Example. The same artery, now with collagen & SMC turning over (elastin does not). Which constituent grows to carry the load, and does turnover restore the set-point? Equilibrium each step: $\bar{\sigma}(\lambda) = \bar{\sigma}_h \, \gamma \, \lambda^2 / m$ (load factor γ, mass ratio m).

Run it.

```
uv run python run.py configs/ex02_constrained_mixture.yaml
```

Sets each constituent's `turnover_days` $(1/k_d)$, `gain` (K_σ), and the `insult`, hypertension (γ) or aneurysm (`elastin_surviving`).

Parametric study.

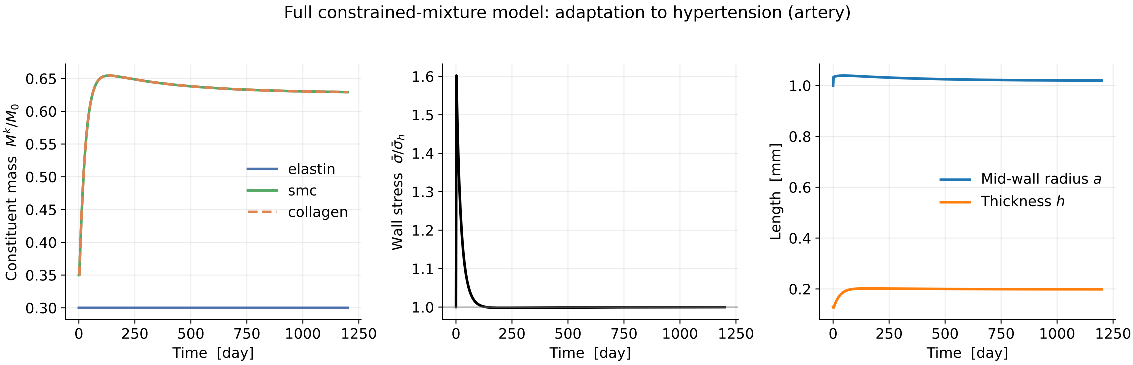
```
...  -set model.constituents.collagen.gain=3.0
```

Your turn: Constrained Mixture Model

Question. Does the same tissue *adapt* to hypertension but *run away* on the aneurysm insult? Which constituent grows to carry the load elastin used to bear?

Answer: Constrained Mixture Model

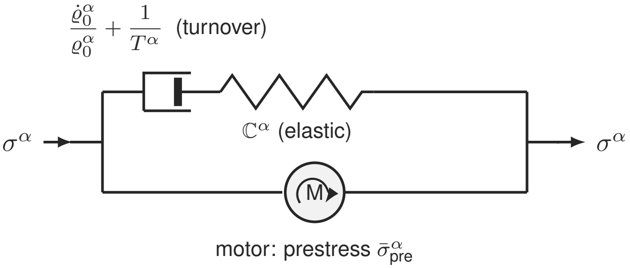
Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover or higher gain reach the new state sooner.



Homogenized CMM

The Mechanical Analog: A Maxwell Fluid with a Motor

Each constituent behaves like a **Maxwell element in parallel with a motor**: turnover relaxes stress toward a *prestress*, not zero.



σ^α	constituent stress (the load)
$\mathbb{C}^\alpha$	elastic stiffness (spring)
$\dot{\varrho}_0^\alpha/\varrho_0^\alpha + 1/T^\alpha$	turnover + degradation (dashpot)
$\bar{\sigma}^\alpha_{\text{pre}}$	deposition prestress (motor)

Back to the full CMM (0D):

- spring $\leftrightarrow$ constituent law $\sigma^\alpha(\lambda_e^\alpha)$
- dashpot $\leftrightarrow$ mass turnover / removal k_d^α
- motor $\leftrightarrow$ deposition stretch G^α

Cyron, Aydin, Humphrey, Biomech Model Mechanobiol (2016)

Growth (Mass) and Remodeling (Natural Configuration)

Two 0D ODEs per constituent replace the cohort history. **Growth**: mass production $\propto$ mass, driven by the tissue-stress deviation. **Remodeling**: new mass enters at natural stretch λ/G^α, old mass leaves at the current λ_n^α; mixing the two at the turnover rate evolves the mean.

0D growth (mass)

$$\frac{\mathrm{d}M^\alpha}{\mathrm{d}t} = M^\alpha K_\sigma^\alpha k_d^\alpha \left(\tfrac{\bar{\sigma}}{\bar{\sigma}_h} - 1\right) \tag{19}$$

0D remodeling (mean natural stretch)

$$\frac{\mathrm{d}\lambda_n^\alpha}{\mathrm{d}t} = \frac{m^\alpha}{M^\alpha}\left(\frac{\lambda}{G^\alpha} - \lambda_n^\alpha\right) \tag{20}$$

with production $m^\alpha = k_d^\alpha M^\alpha \Upsilon^\alpha$. Holding λ fixed, remodeling relaxes each constituent's stress to its prestress σ_h^α (time constant $\sim 1/k_d$), the Maxwell-with-motor behaviour.

Your turn: Why Is the Homogenized Model Cheap?

Question. How does the cost of the homogenized constrained mixture model scale with the number of steps N, compared with the full model?

Your turn: Why Is the Homogenized Model Cheap?

Question. How does the cost of the homogenized constrained mixture model scale with the number of steps N, compared with the full model?

Answer: Why Is the Homogenized Model Cheap?

The state is *fixed-size*, two numbers $(M^\alpha, \lambda_n^\alpha)$ per constituent, and each step just advances two ODEs, with *no* history to re-sum:

$$\text{homogenized: } \mathcal{O}(N) \qquad \text{vs.} \qquad \text{full CMM: } \mathcal{O}(N^2)$$

Same time trajectory, far cheaper, and the fixed state maps onto a viscoelastic (Maxwell-with-motor) material, so it drops traight into standard finite-element solvers.

Your turn: When do homogenized and full agree?

Question. The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model and when would you expect them to diverge?

Your turn: When do homogenized and full agree?

Question. The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model and when would you expect them to diverge?

Answer: When do homogenized and full agree?

They agree when each constituent's cohorts share *nearly the same* natural configuration i.e. slow, gradual loading where deposition stretch G^α and turnover are roughly stationary, so the cohort spread stays narrow and the mean represents it well. **They diverge** under fast or strongly time-varying insults, when cohorts deposited at very different states coexist and a single mean stretch can no longer capture the distribution.

Setup: Example (homogenized vs. full, `ex03`)

Example. Run both mixture theories on the *same* artery and insult. How closely does the homogenized model track the full model, and how much faster is it?

Run it.

```
uv run python run.py configs/ex03_homogenized_vs_full.yaml
```

Here `study: compare` overlays `constrained_mixture`, `homogenized_cmm` and `kinematic_growth` on one scenario, and prints each runtime and the mass-ratio gap between the first two (full vs. homogenized).

Parametric study. Stress the comparison with a harsher, longer insult:

```
...  -set insult.elastin_surviving=0.3 -set insult.ramp=10
...  -set simulate.t_end=4000     # does the agreement hold?
```

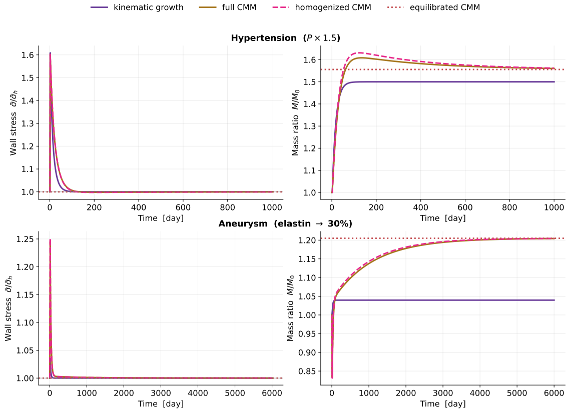
Your turn: Homogenized vs. full

Question. Is the difference between the two curves visible by eye? By what factor is the homogenized model faster and does the agreement survive a harsher, longer insult?

Hints. Read the printed mass-ratio gap and the two runtimes. The homogenized model should track the full one closely wherever the loading is slow and gradual.

Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely at a fraction of the runtime ($O(N)$ vs. $O(N^2)$). **Learning:** it reproduces a **similar time trajectory** to the full model, cheaply.



Equilibrated CMM

Directly Solve for the End State

Set every rate to zero and solve the **algebraic** equilibrium, one root-find instead of thousands of time steps.

3D the 3D equilibrium conditions

$$\underbrace{\Upsilon = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h}_{\text{(A) growth balance}} \quad \underbrace{\boldsymbol{F}_n^\alpha = \text{const} \Rightarrow \sigma^\alpha = \sigma_h^\alpha}_{\text{(B) remodeling completed}} \quad \underbrace{\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}}_{\text{(C) mechanical equilib.}}$$

0D scalar conditions at the evolved geometry

$$\bar{\sigma}^* = \bar{\sigma}_h, \qquad \lambda_e^\alpha = G^\alpha, \qquad \sigma_\theta = \frac{Pa}{h}.$$

Latorre, Humphrey, ZAMM (2018)

Collapse to One Scalar Equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). The 3D conditions (A)–(C) collapse to a *single* equation for the evolved stretch λ^*, consistent with the same $\lambda,\ G^\alpha,\ \phi^\alpha$ used throughout:

0D scalar equilibrium equation for λ^*

$$\phi^e\, s + s\, \frac{\bar{\sigma}_h - \sigma^e(G^e \lambda^*)}{\sigma_{\text{turn}} - \sigma^e_h} - \gamma\, (\lambda^*)^2 = 0 \tag{21}$$

with the Laplace exponent $n = 2$, γ the sustained load factor, s the surviving elastin fraction, ϕ^e the elastin mass fraction. Once λ^* is known, the evolved mass ratio is $M/M_0 = \gamma(\lambda^*)^2$.

This equation *is* the fixed point of the homogenized ODEs ((19)–(20) with $\mathrm{d}/\mathrm{d}t = 0$), so its root coincides with the long-time limit of the transient theory.

The Punchline: Existence = Stability

Equation (21) may have **no physical root**. Then no adapted state exists and the transient theories dilate without bound. The solver is therefore the cleanest **stability test**: root $\Rightarrow$ adapts, no root $\Rightarrow$ unstable. Near the boundary the transient slows critically, which is exactly why the instant algebraic test is so useful.

Setup: Example (the equilibrated stability test, `ex04`)

Example. Skip the transient: solve the single algebraic equation (21) for the *final* adapted stretch λ^*. A physical root means the tissue adapts; *no root* means it dilates without bound. Scanning elastin loss then traces the stability boundary.

Run it.

```
uv run python run.py configs/ex04_equilibrated.yaml
```

Here `study: equilibrium` solves once for the given insult (and prints a transient cross-check: the homogenized plateau lands on λ^*), then the `scan:` block sweeps `elastin_surviving` and `report_critical: elastin` prints the critical surviving fraction s_{crit}.

Parametric study. Move the boundary by changing the wall's load-bearing *capacity*, then re-read s_{crit}:

```
... -set model.constituents.collagen.law.c1=600
... -set model.constituents.collagen.G=1.15
```

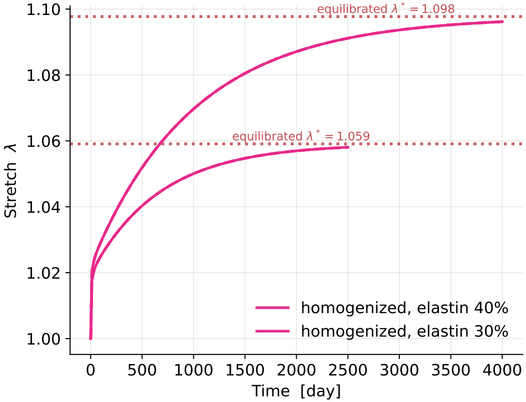
(Raising the `gain` leaves s_{crit} unchanged: gain sets how *fast* the transient adapts, not *whether* an equilibrium exists.)

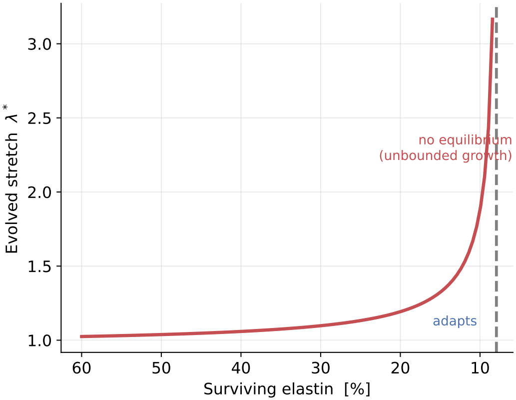
Your turn: Equilibrated

Question. Does the algebraic λ^* land on the transient plateau where a root exists? What is the critical elastin loss beyond which *no* equilibrium exists at all?

Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below s_{crit} there is *no* root: the transient dilates without bound.



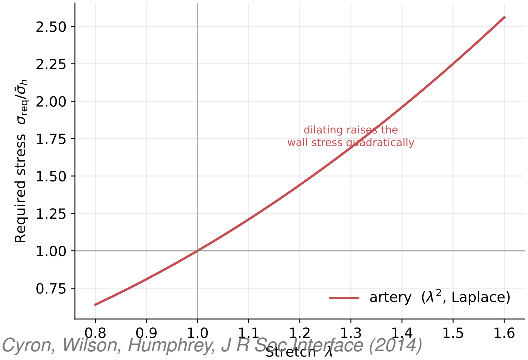


Stability

Where Instability Comes From: The Laplace Exponent

In a pressurised tube the wall stress is $\bar{\sigma} = \frac{Pa}{h}$ with radius $a \propto \lambda$ and thickness $h \propto M$, so the required (Laplace) stress is *quadratic* in the stretch, $\bar{\sigma}_{\text{req}} \propto \gamma\,\lambda^2/M$. The homeostatic state is stable only if adding mass *lowers* the stress it must carry:

$$\left.\frac{\mathrm{d}\bar{\sigma}}{\mathrm{d}\theta}\right|_{\theta^*} < 0 \iff \text{tissue adapts} \tag{22}$$



Growth thickens the wall ($h \propto M$), which lowers stress; but dilation raises it as λ^2 (Laplace). While enough **elastin** survives, its stiff recoil keeps $\mathrm{d}\bar{\sigma}/\mathrm{d}\theta < 0$ and the artery **adapts**. Lose enough elastin and the sign flips, $\mathrm{d}\bar{\sigma}/\mathrm{d}\theta > 0$: **runaway**. It is the quadratic Laplace exponent that makes this possible.

Cyron, Wilson, Humphrey, J R Soc Interface (2014)

Same Tissue, Two Insults

Read stability two ways: **existence**: the equilibrated equation (21) has a root iff the tissue can adapt (no time integration); or **transient**: integrate and watch it plateau or climb. The same tissue can do either: mild elastin loss adapts, severe loss dilates without bound.

Setup: Example (the stable/unstable map, `ex05`)

Example. A tissue *adapts* to an insult iff a mechanobiological equilibrium exists (21). Sweeping that instant test over (surviving elastin) $\times$ (pressure) draws the stable/unstable map. Then change *one* tissue property and watch the unstable "aneurysm" region shrink.

Run it.

```
uv run python run.py configs/ex05_stability.yaml
```

Here `study: map` builds the 2D grid over the two `axes:` via the `adapts` existence test and shades adapt (blue) vs. runaway (red); it also prints the fraction of the grid that adapts.

Parametric study. Enlarge the *adapts* region by giving the wall more load-bearing capacity, then compare the printed fraction:

```
...  -set model.constituents.collagen.law.c1=600
...  -set model.constituents.collagen.G=1.15
```

(Raising the `gain` does *not* move this map: existence is gain-independent. Gain governs *dynamic* stability instead, in the transient.)

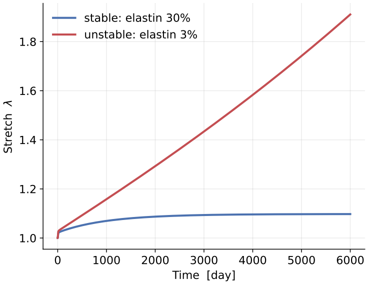
Your turn: Stability map

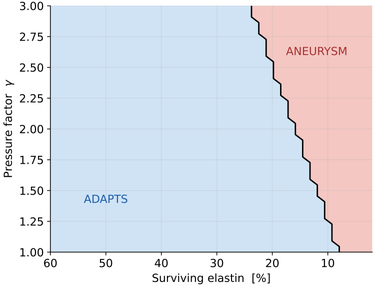
Question. Which capacity change (stiffer collagen or larger deposition stretch) enlarges the stable region the most, and why does raising the `gain` leave the map unchanged? Elastin cannot be regrown in adults: what does that imply for aneurysm?

Hints. Compare the printed "fraction that adapts". Stiffness and deposition stretch move the existence boundary; the gain does not, it sets how fast (and whether) the transient reaches an equilibrium that exists.

Answer: Stability map

Stiffer collagen or a larger deposition stretch enlarge the *adapts* region: the wall carries more load, so it tolerates more elastin loss. The `gain` leaves this existence map unchanged (at equilibrium, production balances removal and the gain cancels); it instead sets *dynamic* stability, whether the transient actually reaches an equilibrium that exists (too weak adaptation runs away even when a root exists).

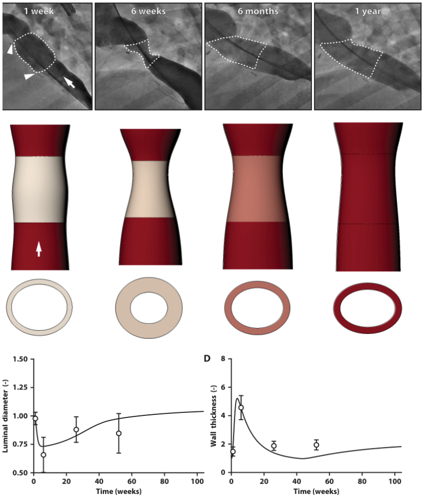




Learning: an equilibrium must *exist* (enough load-bearing capacity) **and** adaptation must be fast enough to *reach* it, existence is necessary, sufficient gain makes it so.

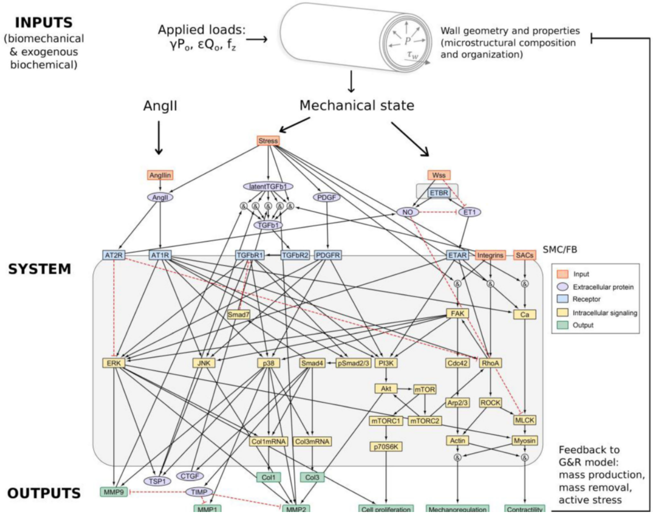
Future Perspectives

Clinical Application: Reversible Graft Stenosis



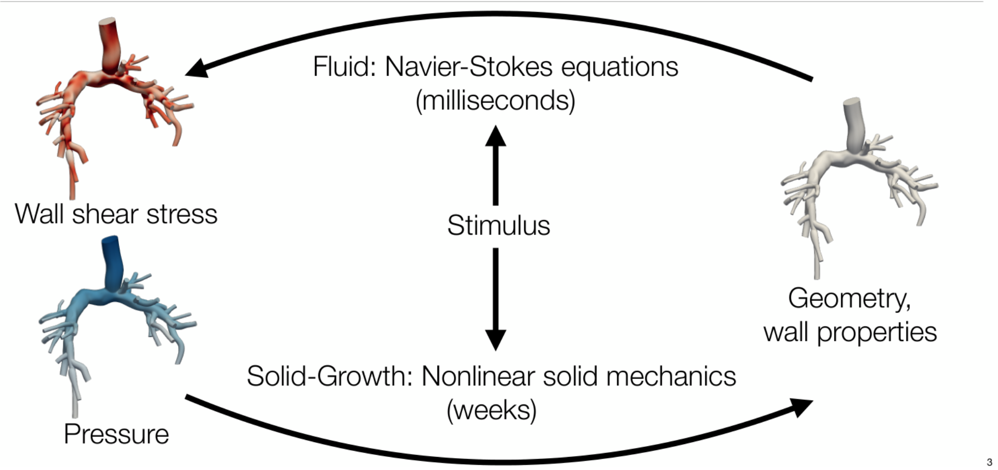
Drews, Pepper, Best, . . . , Humphrey, Breuer, Sci Transl Med (2020)

Systems-Biology Coupling: Signaling to Remodeling



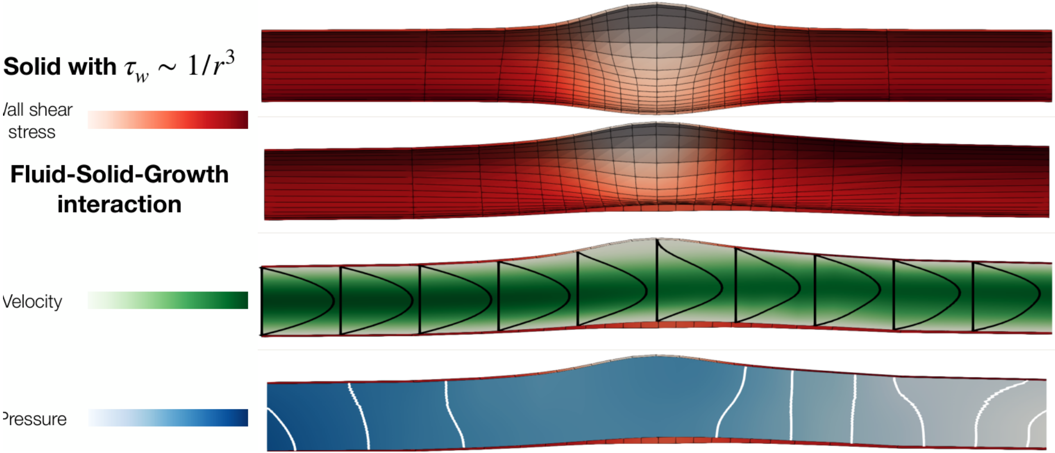
Irons, Humphrey, PLoS Comput Biol (2020); Irons, Latorre, Humphrey, Ann Biomed Eng (2021)

Fluid–Solid–Growth: Closing the Hemodynamic Loop



Schwarz, Pfaller, . . . , Marsden, Comput Methods Appl Mech Eng (2023);
Pfaller, Latorre, . . . , Marsden, Comput Methods Appl Mech Eng (2024)

FSGe: Long-Term Evolved Aneurysmal Equilibrium



Open-Source Implementations

Under the Hood, What Each File Computes (1/2: the core)

The `gr` package is layered: a shared **continuum-mechanics core** (read, don't edit), then one file per G&R theory.

File	Computes	Continuum mechanics / G&R
`mechanics.py`	strain energy $W(\lambda_e)$, 2nd PK stress $S = \mathrm{d}W/\mathrm{d}E_e$, Cauchy $\sigma = \lambda_e^2 S$ (neo-Hookean, Fung)	finite-strain **hyperelasticity** of each constituent, in reference measures
`parameters.py`	`Model`/`Constituent` $(\phi_0, G^\alpha, k_d, K_\sigma)$; set-point $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; `Insult`	**deposition prestretch** & homeostatic **set-points**; the two insults
`geometry.py`	required Laplace stress $\bar{\sigma}_h\,\gamma\,\lambda^2/m$; solves $\bar{\sigma}(\lambda) = \sigma_{\mathrm{req}}$ for λ	**mechanical equilibrium**; no root $\Rightarrow$ runaway
`history.py`	`Result`: $t, \lambda, M/M_0, \bar{\sigma}, \ldots$	shared time-history output

Under the Hood, What Each File Computes (2/2: the theories)

File	Computes	G&R theory / role
`kinematic_growth.py`	split $\boldsymbol{F} = \boldsymbol{F}_e \boldsymbol{F}_g$, scalar θ; $\dot{\theta} = k_g \theta(\bar{\sigma}/\bar{\sigma}_h - 1)$	Theory 1: kinematic growth
`constrained_mixture.py`	heredity integrals over cohorts; mass balance + mixture stress ($O(N^2)$)	Theory 2: full CMM
`homogenized_cmm.py`	two ODEs per constituent: $\dot{M}^\alpha$, $\dot{\lambda}_n^\alpha$ ($O(N)$)	Theory 3: homogenized CMM
`equilibrated_cmm.py`	rates $\to 0$: one algebraic solve for λ^*	Theory 4: equilibrated CMM
`stability.py`	`adapts` (does a root exist?), `critical_elastin_loss`	stability $=$ **existence** of an equilibrium
`scenario.py`, `run.py`, `configs/`	build model/geometry/insult from YAML; run a study; sweep	the exercise driver (input files)

Common thread: each theory changes only *how the supplied stress* $\bar{\sigma}(\lambda)$ *is built*, the same constituent laws and equilibrium solve are reused, so every difference you saw is the G&R law alone.

Research-Grade Open-Source Solvers

Project	What it does	Stack / group
svGrowth	constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above	Python; Stanford CBCL
svFSGe	fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)	Python + svFSI/svMultiPhysics; Stanford CBCL
FEMBE_Plugin	mechanobiologically equilibrated constrained-mixture material as a FEBio plugin	C++ / FEBio; Yale (Humphrey lab)
FEFSG_Plugin	fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)	C++ / FEBio; Yale (Humphrey lab)

`github.com/StanfordCBCL/svGrowth` `github.com/StanfordCBCL/svFSGe`

`github.com/yale-humphrey-lab/FEMBE_Plugin` `github.com/yale-humphrey-lab/FEFSG_Plugin`

svGrowth builds on Humphrey, Rajagopal, Math Models Methods Appl Sci (2002); FEMBE on Latorre, Humphrey, Comput Methods Appl Mech Eng (2020)

How the Four Theories Answer: The Takeaways

Theory	Stability?	What it says
Kinematic growth	*Imposed*	Either reaches the *a-priori prescribed* homeostasis, or is unstable, nothing in between. Built in, not learned.
Constrained mixture (full)	*Predicted*	Adapts or dilates **depending on the insult**, the faithful reference.
Homogenized CMM	*Predicted*	A **similar time trajectory** to the full model, at a fraction of the cost.
Equilibrated CMM	*Instantly*	**Matches** the transient theories *if* an equilibrium exists; *no* root $\Rightarrow$ it does not match, unbounded growth.

Summary: What to Take Away

The biology. Living tissue continuously **turns over**; cells deposit new material *pre-stretched* at G^α, chasing a homeostatic mechanical set-point σ_h.

The four theories, one set-point, one question: does it adapt or run away?

- **Kinematic growth**: prescribe $\boldsymbol{F}_g$; *imposes* the set-point. Outcome is binary: reach the prescribed homeostasis, or go unstable.
- **Constrained mixture**: track every cohort; stability is *predicted* and **depends on the insult** (mild $\rightarrow$ adapt, severe $\rightarrow$ runaway).
- **Homogenized**: one mean natural configuration per constituent; a **similar time trajectory** to the full model at $O(N)$ not $O(N^2)$.
- **Equilibrated**: solve for the end state directly; **matches** the transient theories *iff* an equilibrium exists, and flags instability by finding no root.

Where it goes. These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

Key references I

- Rodriguez, Hoger, McCulloch. Stress-dependent finite growth in soft elastic tissues. *J Biomech* (1994).
- Humphrey, Rajagopal. A constrained mixture model for growth and remodeling of soft tissues. *Math Models Methods Appl Sci* (2002).
- Cyron, Wilson, Humphrey. Mechanobiological stability. *J R Soc Interface* (2014).
- Cyron, Aydin, Humphrey. A homogenized constrained mixture (and mechanical analog) model. *Biomech Model Mechanobiol* (2016).
- Latorre, Humphrey. A mechanobiologically equilibrated constrained mixture model. *ZAMM* (2018).
- Latorre, Humphrey. Fast, rate-independent finite-element implementation of a 3D constrained mixture model. *Comput Methods Appl Mech Eng* (2020).
- Humphrey. Constrained mixture models of soft tissue growth and remodeling. *J Elasticity* (2021).
- Irons, Humphrey. Cell signaling model for arterial mechanobiology. *PLoS Comput Biol* (2020).
- Irons, Latorre, Humphrey. From transcript to tissue: multiscale modeling. *Ann Biomed Eng* (2021).
- Drews, Pepper, Best, . . . , Humphrey, Breuer. Spontaneous reversal of stenosis in tissue-engineered vascular grafts. *Sci Transl Med* (2020).

Key references II

- Figueroa, Baek, Taylor, Humphrey. A computational framework for fluid–solid–growth modeling. *Comput Methods Appl Mech Eng* (2009).
- Schwarz, Pfaller, . . . , Marsden. A fluid–solid–growth solver for cardiovascular modeling. *Comput Methods Appl Mech Eng* (2023).
- Pfaller, Latorre, . . . , Marsden. FSGe: a fast, strongly coupled 3D fluid–solid–growth method. *Comput Methods Appl Mech Eng* (2024).
- Humphrey, Eberth, Dye, Gleason. Fundamental role of axial stress in arterial mechanics. *J Biomech* (2009).
- Ambrosi, Ben Amar, Cyron, . . . , Humphrey, Kuhl. Growth and remodelling of living tissues. *J R Soc Interface* (2019).

Key references III

Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

Figure sources. Result figures reproduced with the `gr` package; literature figures cited per-slide.